Name – Aditi Sawarkar

PRN – 24070521118

Semester – 4th

Section – B

Title : Student Management System using Inheritance in Java

Problem Definition

Design a Java program to manage student information using inheritance.
Create a base class Person that stores common details like name and age.
Derive a class Student that includes additional attributes such as roll number and marks.
Further extend it into another class GraduateStudent with specialization details.
The program should display complete student information using inherited properties.

Objective:

- To understand inheritance in Java
- To implement multi-level inheritance
- To reuse code using base and derived classes
- To display student details using hierarchical structure

Theory:

Inheritance is an important concept of Object-Oriented Programming (OOP).
It allows one class to acquire properties and methods of another class.

Types used in this program:

- Single Inheritance → Student inherits Person
- Multilevel Inheritance → GraduateStudent inherits Student

Advantages:

- Code reusability
- Easy maintenance
- Improves readability

Algorithm:

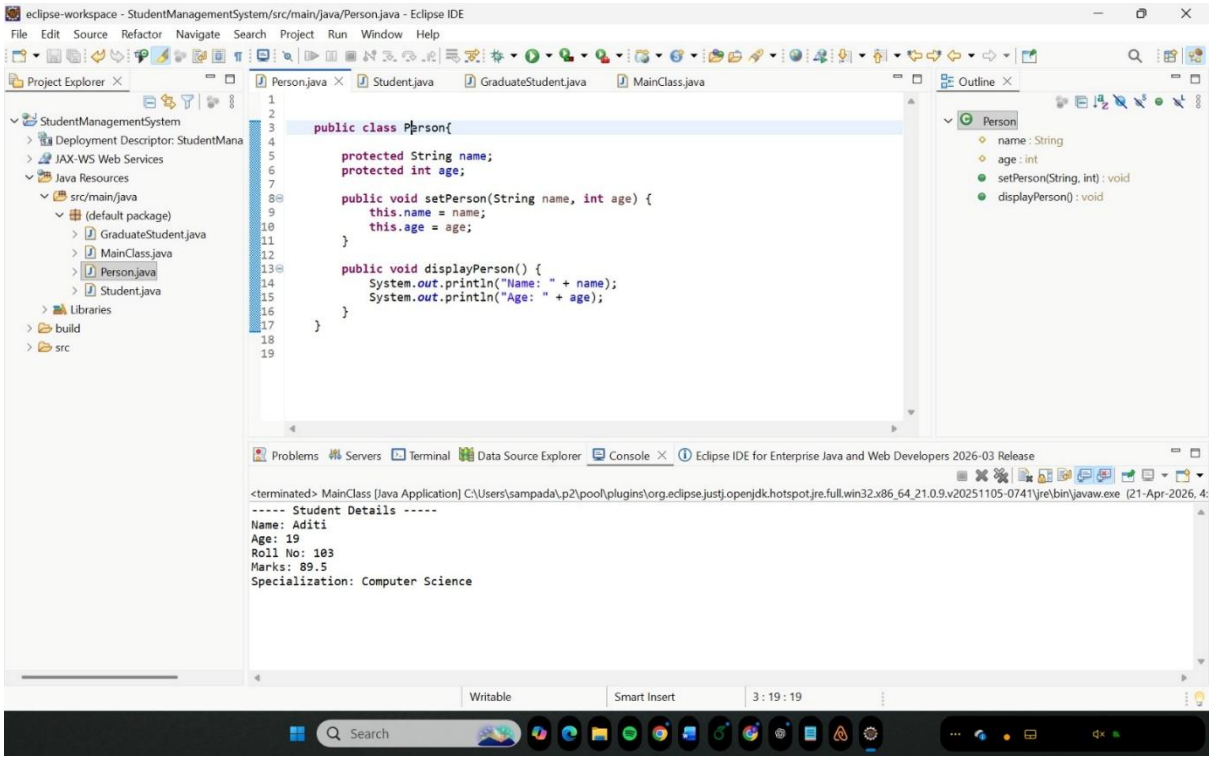
1. Start the program

2. Create class Person with name and age
3. Create class Student inheriting Person with rollNo and marks
4. Create class GraduateStudent inheriting Student with specialization
5. Create object of GraduateStudent
6. Set all values using methods
7. Display all details
8. End program

1.Person.java

```
public class Person {
    protected String name;
    protected int age;
    public void setPerson(String name, int age) {
        this.name = name;
        this.age = age;
    }
    public void displayPerson() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}
```

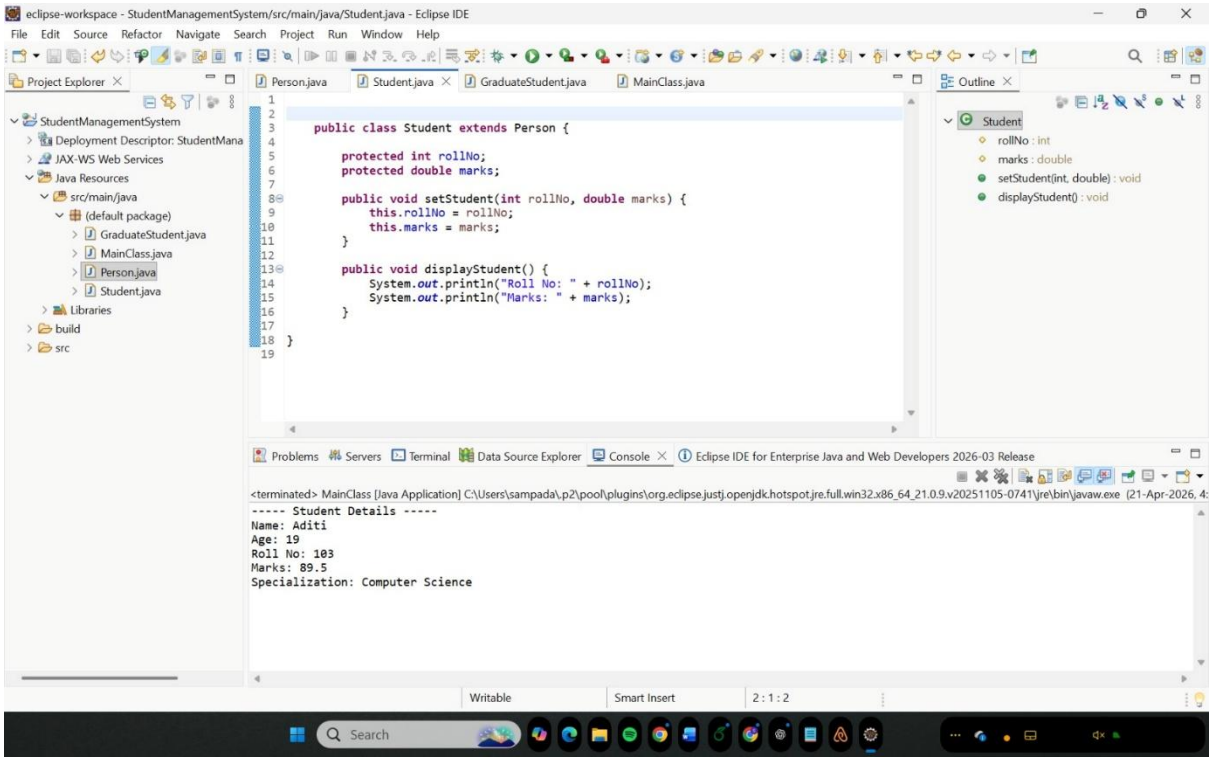
Output:



2.Student.java

```
public class Student extends Person {
    protected int rollNo;
    protected double marks;
    public void setStudent(int rollNo, double marks) {
        this.rollNo = rollNo;
        this.marks = marks;
    }
    public void displayStudent() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Marks: " + marks);
    }
}
```

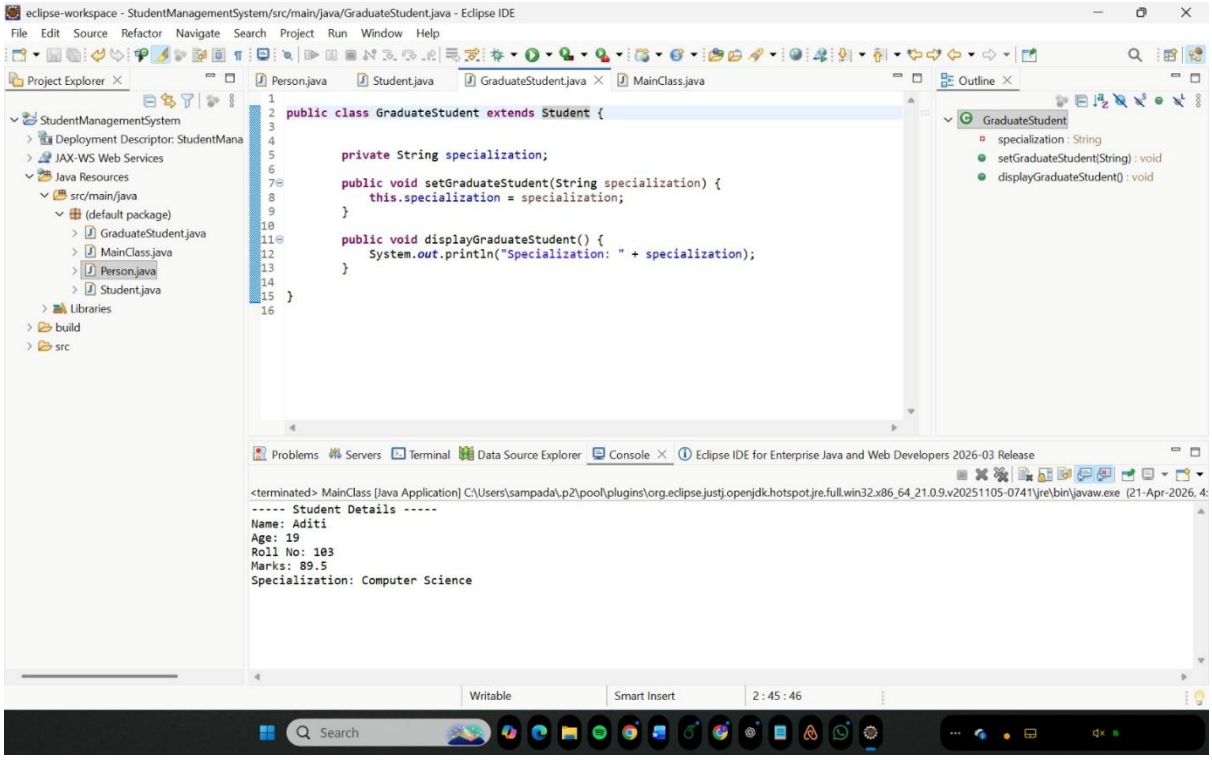
Output:



3.GraduateStudent.java

```
public class GraduateStudent extends Student {
    private String specialization;
    public void setGraduateStudent(String specialization) {
        this.specialization = specialization;
    }
    public void displayGraduateStudent() {
        System.out.println("Specialization: " + specialization);
    }
}
```

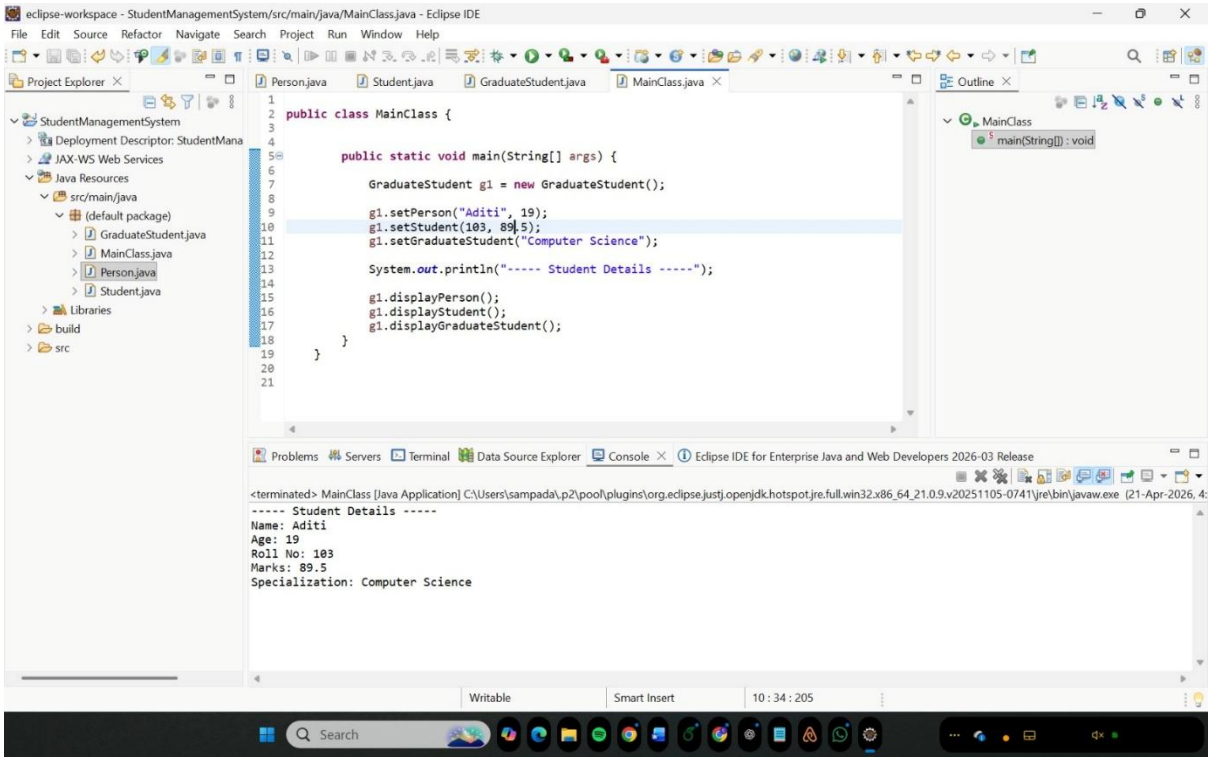
Output:



4.MainClass.java

```
public class MainClass {
    public static void main(String[] args) {
        GraduateStudent g1 = new GraduateStudent();
        g1.setPerson("Aditi", 19);
        g1.setStudent(103, 89.5);
        g1.setGraduateStudent("Computer Science");
        System.out.println("----- Student Details -----");
        g1.displayPerson();
        g1.displayStudent();
        g1.displayGraduateStudent();
    }
}
```

Output:



Conclusion

The program successfully demonstrates the concept of multilevel inheritance in Java.
It shows how data can be reused across multiple classes and how a derived class can access properties of its parent classes.
This approach improves code structure and reduces redundancy.